SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO2: Apply the relational model, relational algebra, SQL query structures, and views

to solve database querying and data manipulation problems. (BL3, BL6)

Activity Tool : Analytical Problem Solving (High)

Assessment Tool Weightage: 40%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Given a set of relations and sample data, write SQL queries to retrieve specific information using joins and subqueries, and explain the logic used. Bloom's Level: BL3 – Apply	BL4 – Analyze
2	Using relational algebra, formulate expressions to solve complex queries such as retrieving records that satisfy multiple conditions across relations. Bloom's Level: BL4 – Analyse	BL5 – Evaluate
3	Given a real-world scenario (e.g., banking or student system), design a relational schema and construct appropriate SQL queries for data retrieval and updates. Bloom's Level: BL6 – Create	BL6 – Create
4	Analyse a given SQL query with nested subqueries and predict its output, explaining each step of execution. Bloom's Level: BL4 – Analyze	BL6 – Create

5	**Compare different SQL approaches (joins vs subqueries) for solving the same problem and evaluate their efficiency.** **Bloom's Level: BL5 – Evaluate**	**BL4 – Analyze**
6	**Design and implement views for a database system to provide restricted data access, and justify their use for security and abstraction.** **Bloom's Level: BL6 – Create**	**BL5 – Evaluate**
7	**Given a database schema, write relational algebra expressions and convert them into equivalent SQL queries, analyzing differences in representation.** **Bloom's Level: BL4 – Analyze**	**BL4 – Analyze**
8	**Optimize a set of inefficient SQL queries by restructuring them using joins, indexing concepts, or views, and evaluate performance improvements.** **Bloom's Level: BL5 – Evaluate**	**BL5 – Evaluate**
9	**Develop a complex SQL query involving grouping, aggregation, and nested queries to solve a real-world problem.** **Bloom's Level: BL6 – Create**	**BL6 – Create**
10	**Given multiple relations, analyze the query requirements and decide whether to use views, joins, or subqueries, justifying your choice with reasoning.** **Bloom's Level: BL5 – Evaluate**	**BL5 – Evaluate**

Question 1

Given a set of relations and sample data, write SQL queries to retrieve specific information using joins and subqueries, and explain the logic used. Bloom's Level: BL3 – Apply

Query 1

```
mysql> CREATE DATABASE DBMS_AT1;
Query OK, 1 row affected (0.02 sec)

mysql> USE DBMS_AT1;
Database changed
mysql> CREATE TABLE Department(
    -> DeptID INT PRIMARY KEY,
    -> DeptName VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE Student(
    -> SID INT PRIMARY KEY,
    -> Name VARCHAR(30),
    -> DeptID INT,
    -> Marks INT,
    -> FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Department VALUES
    -> (1,'CSE'),
    -> (2,'ECE'),
    -> (3,'EEE');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Student VALUES
    -> (101,'Rahul',1,85),
    -> (102,'Anita',2,92),
    -> (103,'Kiran',1,75),
    -> (104,'Sneha',3,88),
    -> (105,'Ravi',2,67),
    -> (106,'Asha',1,95);
Query OK, 6 rows affected (0.03 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT S.SID,S.Name,D.DeptName
    -> FROM Student S
    -> JOIN Department D
    -> ON S.DeptID=D.DeptID;
+-----+--------+----------+
| SID | Name   | DeptName |
+-----+--------+----------+
| 101 | Rahul  | CSE      |
| 103 | Kiran  | CSE      |
| 106 | Asha   | CSE      |
| 102 | Anita  | ECE      |
| 105 | Ravi   | ECE      |
| 104 | Sneha  | EEE      |
+-----+--------+----------+
6 rows in set (0.02 sec)
```

Query 2

```
mysql> SELECT S.Name, D.DeptName
    -> FROM Student S
    -> JOIN Department D
    -> ON S.DeptID = D.DeptID;
+--------+----------+
| Name   | DeptName |
+--------+----------+
| Rahul  | CSE      |
| Kiran  | CSE      |
| Asha   | CSE      |
| Anita  | ECE      |
| Ravi   | ECE      |
| Farhan | ECE      |
| Sneha  | EEE      |
+--------+----------+
7 rows in set (0.00 sec)
```

Query 3

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE DeptID =
    -> (
    -> SELECT DeptID
    -> FROM Department
    -> WHERE DeptName='CSE'
    -> );
+-------+
| Name  |
+-------+
| Rahul |
| Kiran |
| Asha  |
+-------+
3 rows in set (0.00 sec)
```

Query 4

```
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks >
    -> (
    -> SELECT AVG(Marks)
    -> FROM Student
    -> );
+--------+-------+
| Name   | Marks |
+--------+-------+
| Anita  |    92 |
| Sneha  |    88 |
| Asha   |    95 |
| Farhan |    95 |
+--------+-------+
4 rows in set (0.00 sec)
```

Question 2

Using relational algebra, formulate expressions to solve complex queries such as retrieving records that satisfy multiple conditions across relations.
Bloom's Level: BL4 – Analyse

Relational algebra

1. Retrieve all CSE students

$\pi_{Name} (\sigma_{DeptName='CSE'} (Student \bowtie Department))$

2. Retrieve students with Marks > 80

$\pi_{SID,\ Name} (\sigma_{Marks>80} (Student))$

3. Retrieve ECE students with Marks > 70

$\pi_{Name} (\sigma_{DeptName='ECE' \wedge Marks>70} (Student \bowtie Department))$

Question 3

Given a real-world scenario (e.g., banking or student system), design a relational schema and construct appropriate SQL queries for data retrieval and updates. Bloom's Level: BL6 – Create

Query 1

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE DeptID=
    -> (
    -> SELECT DeptID
    -> FROM Department
    -> WHERE DeptName='CSE'
    -> );
+--------+
| Name   |
+--------+
| Rahul  |
| Kiran  |
| Asha   |
+--------+
3 rows in set (0.02 sec)
```

Query 2

```
mysql> SELECT *
    -> FROM Student
    -> WHERE DeptID=1
    -> AND Marks>80;
+-----+-------+--------+-------+
| SID | Name  | DeptID | Marks |
+-----+-------+--------+-------+
| 101 | Rahul |      1 |    85 |
| 106 | Asha  |      1 |    95 |
+-----+-------+--------+-------+
2 rows in set (0.02 sec)
```

Query 3

```
mysql> SELECT * FROM Student;
+-----+-------+--------+-------+
| SID | Name  | DeptID | Marks |
+-----+-------+--------+-------+
| 101 | Rahul |      1 |    85 |
| 102 | Anita |      2 |    92 |
| 103 | Kiran |      1 |    75 |
| 104 | Sneha |      3 |    88 |
| 105 | Ravi  |      2 |    67 |
| 106 | Asha  |      1 |    95 |
+-----+-------+--------+-------+
6 rows in set (0.02 sec)
```

Question 4

Analyse a given SQL query with nested subqueries and predict its output, explaining each step of execution. Bloom's Level: BL4 – Analyze

Query 1

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE Marks=
    -> (
    -> SELECT MAX(Marks)
    -> FROM Student
    -> );
+--------+
| Name   |
+--------+
| Asha   |
| Farhan |
+--------+
2 rows in set (0.00 sec)
```

Query 2

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE DeptID=
    -> (
    -> SELECT DeptID
    -> FROM Department
    -> WHERE DeptName='ECE'
    -> );
+--------+
| Name   |
+--------+
| Anita  |
| Ravi   |
| Farhan |
+--------+
3 rows in set (0.00 sec)
```

Query 3

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE Marks>
    -> (
    -> SELECT AVG(Marks)
    -> FROM Student
    -> );
+--------+
| Name   |
+--------+
| Anita  |
| Sneha  |
| Asha   |
| Farhan |
+--------+
4 rows in set (0.00 sec)
```

Question 5

Compare different SQL approaches (joins vs subqueries) for solving the same problem and evaluate their efficiency. Bloom's Level: BL5 – Evaluate

Query 1

```
mysql> SELECT Name,DeptName
    -> FROM Student
    -> JOIN Department
    -> ON Student.DeptID=Department.DeptID;
+--------+----------+
| Name   | DeptName |
+--------+----------+
| Rahul  | CSE      |
| Kiran  | CSE      |
| Asha   | CSE      |
| Anita  | ECE      |
| Ravi   | ECE      |
| Farhan | ECE      |
| Sneha  | EEE      |
+--------+----------+
7 rows in set (0.00 sec)
```

Query 2

```
mysql> SELECT Name,
    -> (
    ->     SELECT DeptName
    ->     FROM Department
    ->     WHERE Department.DeptID = Student.DeptID
    -> ) AS Department
    -> FROM Student;
+--------+------------+
| Name   | Department |
+--------+------------+
| Rahul  | CSE        |
| Anita  | ECE        |
| Kiran  | CSE        |
| Sneha  | EEE        |
| Ravi   | ECE        |
| Asha   | CSE        |
| Farhan | ECE        |
+--------+------------+
7 rows in set (0.00 sec)
```

Query 3

```
mysql> SELECT *
    -> FROM Student
    -> LEFT JOIN Department
    -> ON Student.DeptID=Department.DeptID;
+-----+--------+--------+-------+--------+----------+
| SID | Name   | DeptID | Marks | DeptID | DeptName |
+-----+--------+--------+-------+--------+----------+
| 101 | Rahul  |      1 |    85 |      1 | CSE      |
| 102 | Anita  |      2 |    92 |      2 | ECE      |
| 103 | Kiran  |      1 |    75 |      1 | CSE      |
| 104 | Sneha  |      3 |    88 |      3 | EEE      |
| 105 | Ravi   |      2 |    67 |      2 | ECE      |
| 106 | Asha   |      1 |    95 |      1 | CSE      |
| 107 | Farhan |      2 |    95 |      2 | ECE      |
+-----+--------+--------+-------+--------+----------+
7 rows in set (0.02 sec)
```

Question 6

Design and implement views for a database system to provide restricted data access, and justify their use for security and abstraction. Bloom's Level: BL6 – Create

Query

```
mysql> CREATE VIEW CSE_Students AS
    -> SELECT SID,Name,Marks
    -> FROM Student
    -> WHERE DeptID=1;
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM CSE_Students;
+-----+-------+-------+
| SID | Name  | Marks |
+-----+-------+-------+
| 101 | Rahul |    85 |
| 103 | Kiran |    75 |
| 106 | Asha  |    95 |
+-----+-------+-------+
3 rows in set (0.03 sec)
```

Question 7

Given a database schema, write relational algebra expressions and convert them into equivalent SQL queries, analyzing differences in representation. Bloom's Level: BL4 – Analyze

Expression 1

πDeptName (Department)

Expression 2

πName, Marks (σDeptName='CSE' (Student $\bowtie$ Department))

Expression 3

πDeptName (σMarks>90 (Student $\bowtie$ Department))

Question 8

Optimize a set of inefficient SQL queries by restructuring them using joins, indexing concepts, or views, and evaluate performance improvements. Bloom's Level: BL5 – Evaluate

Query 1

```
mysql> SELECT *
    -> FROM Student
    -> WHERE DeptID IN
    -> (
    -> SELECT DeptID
    -> FROM Department
    -> WHERE DeptName='CSE'
    -> );
+-----+-------+--------+-------+
| SID | Name  | DeptID | Marks |
+-----+-------+--------+-------+
| 101 | Rahul |      1 |    85 |
| 103 | Kiran |      1 |    75 |
| 106 | Asha  |      1 |    95 |
+-----+-------+--------+-------+
3 rows in set (0.00 sec)
```

Query 2

```
mysql> SELECT S.*
    -> FROM Student S
    -> JOIN Department D
    -> ON S.DeptID=D.DeptID
    -> WHERE D.DeptName='CSE';
+-----+-------+--------+-------+
| SID | Name  | DeptID | Marks |
+-----+-------+--------+-------+
| 101 | Rahul |      1 |    85 |
| 103 | Kiran |      1 |    75 |
| 106 | Asha  |      1 |    95 |
+-----+-------+--------+-------+
3 rows in set (0.00 sec)
```

Question 9

Develop a complex SQL query involving grouping, aggregation, and nested queries to solve a real-world problem. Bloom's Level: BL6 – Create

Query 1

```
mysql> SELECT DeptID,
    -> COUNT(*) AS Total
    -> FROM Student
    -> GROUP BY DeptID;
+--------+-------+
| DeptID | Total |
+--------+-------+
|      1 |     3 |
|      2 |     3 |
|      3 |     1 |
+--------+-------+
3 rows in set (0.00 sec)
```

Query 2

```
mysql> SELECT DeptID,
    -> AVG(Marks)
    -> FROM Student
    -> GROUP BY DeptID;
+--------+------------+
| DeptID | AVG(Marks) |
+--------+------------+
|      1 |    85.0000 |
|      2 |    84.6667 |
|      3 |    88.0000 |
+--------+------------+
3 rows in set (0.00 sec)
```

Query 3

```
mysql> SELECT DeptID,
    -> MAX(Marks)
    -> FROM Student
    -> GROUP BY DeptID;
+--------+------------+
| DeptID | MAX(Marks) |
+--------+------------+
|      1 |         95 |
|      2 |         95 |
|      3 |         88 |
+--------+------------+
3 rows in set (0.00 sec)
```

Question 10

Given multiple relations, analyze the query requirements and decide whether to use views, joins, or subqueries, justifying your choice with reasoning. Bloom's Level: BL5 – Evaluate

Query 1

```
mysql> SELECT Name,DeptName
    -> FROM Student
    -> JOIN Department
    -> ON Student.DeptID=Department.DeptID;
+--------+----------+
| Name   | DeptName |
+--------+----------+
| Rahul  | CSE      |
| Kiran  | CSE      |
| Asha   | CSE      |
| Anita  | ECE      |
| Ravi   | ECE      |
| Farhan | ECE      |
| Sneha  | EEE      |
+--------+----------+
7 rows in set (0.00 sec)
```

Query 2

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE DeptID=
    -> (
    -> SELECT DeptID
    -> FROM Department
    -> WHERE DeptName='EEE'
    -> );
+-------+
| Name  |
+-------+
| Sneha |
+-------+
1 row in set (0.00 sec)
```

Code of Conduct:

I _A Mohammad Naheed-192525336_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **A Mohammad Naheed**
192525336